

Software Requirements Specification (SRS)

Project Title: Examination System Database

1. Introduction

1.1 Purpose

The purpose of this system is to automate the academic examination process, including question bank management, random exam generation, student enrollment, and automated grading for an educational institution (like ITI).

1.2 Scope

The system handles the storage of organizational structures (Branches, Tracks), user management (Instructors, Students, Managers), and the full lifecycle of an exam—from question creation to result analysis.

2. User Roles & Permissions (Security Model)

The system implements Role-Based Access Control (RBAC) via SQL Server Logins and Users:

Role	Permissions & Responsibilities
System Admin	Full database access (db_owner). Responsible for maintenance and backups.
Training Manager	Manage infrastructure (Branches, Tracks, Intakes). Add/Edit student and instructor profiles.
Instructor	Manage question pools for assigned courses. Create and schedule exams. Review student answers.
Student	Access assigned exams during scheduled windows. Submit answers and view results.

۲. Functional Requirements

۲,۱ Infrastructure Management

- The system shall allow Training Managers to define Branches, Departments, Tracks, and Intakes.
- Students must be mapped to a specific Branch, Track, and Intake.

۲,۲ Question Bank Management

- Instructors can add questions of three types: **MCQ**, **True/False**, and **Text**.
- Each question must be associated with a specific Course.
- Correct answers (or "Best Answers" for text) must be stored for automated/assisted grading.

۲,۳ Exam Generation & Scheduling

- Instructors can generate exams by specifying the number of questions per type.
- The system shall select questions **randomly** from the pool.
- **Integrity Rule:** The total degrees of questions in an exam must not exceed the Course's MaxDegree.
- Exams must have a defined StartTime and EndTime. Students cannot access exams outside this window.

۲,۴ Automated Grading & Results

- MCQ and T/F questions are graded automatically by a system trigger upon submission.
 - Text-based questions use string-matching functions to provide a suggested score for instructor review.
 - The system calculates a final percentage and a letter grade (A, B, C, D, F) based on the total result.
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4. Technical Requirements

4.1 Data Storage (Filegroups)

To optimize performance, the database is partitioned into:

- **Primary:** System metadata.
- **StaticData:** Rarely changed info (Roles, Branches, Courses).
- **TransactionalData:** High-frequency data (Users, Questions, Student Answers).
- **Indexes:** Dedicated filegroup for index storage to speed up search queries.

4.2 Performance & Optimization

- **Non-Clustered Indexes** are implemented on Email and CourseID columns to ensure fast login and exam retrieval.
- **Views** are used to provide pre-calculated reports (e.g., V_ExamAnalytics) to reduce complex join overhead for end-users.

4.3 Data Integrity

- **Triggers:** Used to enforce complex business rules (e.g., preventing over-scoring an exam).
- **Constraints:** Check constraints ensure valid data types and logical time sequences (Start < End).

5. Non-Functional Requirements

5.1 Security

- Password-protected logins for all users.
- Encryption on sensitive views (e.g., VW_StudentAnswer) to hide grading logic.

5.2 Availability & Backup

- The system requires a **Daily Automatic Backup** to prevent data loss in the transactional environment.